

Database Project Proposal

1) Which of the 17 SDGs will you target?

This project will specifically target Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure.

2) What question will you attempt to answer?

The primary research question is: **What technology infrastructure improvements are most needed in underfunded schools to provide students with reliable access to technology for their education?**

We want to look at problems such as schools not having enough computers or tablets, outdated devices, slow internet, unreliable Wi-Fi, and not having enough money in the budget to make technology improvements.

3) Where will the data come from for this solution?

The data will come from a survey sent to teachers, school administrators, and IT staff at underfunded schools. We would use the survey to find out what technology each school currently has and what areas need the most improvement.

Some questions on the survey would include:

1. How many students attend your school?
2. How many computers or tablets are available for students?
3. About how old are the computers or tablets?
4. How many students typically have to share one device?
5. What is the average internet speed at the school?
6. How often does the school experience internet or Wi-Fi problems each month?
7. Does every classroom have reliable Wi-Fi?
8. What technology issue causes the most problems for students and teachers?
9. What technology improvement does your school need the most?
10. About how much does your school spend on technology each year?

We would also ask the schools to rank what they need most from the following choices: new computers, tablets, faster internet, better Wi-Fi, smartboards, computer labs, and more IT support.

4) Proposed Data Structure / Schema

Column Name	Data Type	Example
SchoolID	INT	1
SchoolName	VARCHAR(100)	Valley High School
Location	VARCHAR(100)	Newburgh, NY
NumberOfStudents	INT	500
NumberOfDevices	INT	125
StudentsPerDevice	DECIMAL(4, 1)	4.0
AverageDeviceAge	DECIMAL(3, 1)	6.0
InternetSpeedMbps	DECIMAL(8, 2)	50.00
WiFiProblemsPerMonth	INT	12
AnnualTechnologyBudget	DECIMAL(10, 2)	20000.00
GreatestTechnologyNeed	VARCHAR(100)	New computers